
EEG-RQVAE: Semantic EEG Representations via Residual Quantization and Latent Predictive Pretraining

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Abstract

Electroencephalography (EEG) foundation models are predominantly pretrained by reconstructing the raw signal, an objective that spends model capacity on low-level noise and physiological artefacts rather than on the semantic structure useful for downstream brain–computer interfaces (BCIs). Moreover, existing models rarely produce discrete, reusable token representations, and none combine residual vector-quantized tokenization with a latent self-supervised objective. We introduce **EEG-RQVAE**, a residual quantized variational autoencoder that learns hierarchical discrete EEG tokens through a two-stage pipeline: a denoising adversarial reconstruction stage that builds a compact codebook, followed by a JEPA-style stage that refines the encoder purely in latent space, without any signal-level reconstruction pressure. A linear-time Mamba backbone models long, multi-channel recordings efficiently. Under a unified, fully subject-independent linear-probing benchmark spanning twelve BCI datasets, a frozen RVQ-Mamba encoder is the most consistent variant—best or second-best on every evaluated task—and improves over recent EEG foundation models on motor-imagery (BCI IV-2a), emotion (SEED-V), pathology (TUAB) and sleep-staging (Sleep-EDF) tasks.

Code : <https://github.com/yrothlin-03/EEG-VAE>

Note: This paper is not publishable in its current form. Several baselines and ablations are still missing (currently being run). Furthermore, the pretraining scale (Stages 1 and 2) is too limited to draw strong conclusions or genuinely claim to provide a foundation model—scaling it up requires substantial computational resources and code optimization that I did not have time to complete. Finally, although Stage 2—arguably the most notable contribution—is designed to build a more semantic latent space, the current results do not yet strongly support this claim. A more thorough analysis of the codebook is still needed to confirm that the tokens learned after Stage 2 are indeed more semantic and encode meaningful EEG features.

1. Introduction

Electroencephalography (EEG) measures the brain’s electrical activity through electrodes placed on the scalp, capturing oscillatory dynamics across a broad frequency spectrum with millisecond temporal resolution (Teplan, 2002; Müller-Putz, 2020). These properties make EEG a central modality in brain-computer interfaces (BCIs), with applications spanning motor imagery (Pfurtscheller & Neuper, 2001), imagined speech decoding (Elwasify et al., 2026), cognitive monitoring, and clinical diagnosis. Yet EEG signals are inherently noisy, non-stationary, and highly subject-specific, which makes decoding a notoriously difficult problem. Traditional approaches relied on hand-crafted features and shallow classifiers, but deep learning has substantially improved performance across tasks (Roy et al., 2019). Despite this progress, most methods remain task-specific and require large labeled datasets for each new application—a bottleneck that severely limits their clinical and real-world deployment.

The success of large pretrained models in natural language processing and computer vision has inspired a growing body of work on *EEG foundation models*—models trained on diverse, unlabeled EEG corpora and then adapted to downstream tasks via lightweight probing (Alain & Bengio, 2017). Several such models have been proposed recently, leveraging the Transformer architecture (Vaswani et al., 2017): EEGPT (Wang et al., 2024), CBraMod (Wang et al., 2025a), REVE (Ouahidi

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et al., 2026), and LUNA (Döner et al., 2025) adopt masked reconstruction as the pretraining objective, while CSBrain (Zhou et al., 2026) introduces a cross-scale spatiotemporal approach. In parallel, state space models (SSMs) such as Mamba (Gu & Dao, 2024) and Mamba-2 (Dao & Gu, 2024) have emerged as efficient alternatives to attention-based architectures, with linear complexity in sequence length—a desirable property for long EEG recordings. EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), and LuMamba (Broustail et al., 2026) explore this direction, but their pretraining objectives remain anchored to pixel-space reconstruction in the input signal domain. Concurrently, BAM-U-Net (Author, s) combines BiMamba-2 blocks with an adversarial denoising objective and achieves strong linear probing performance; however, it operates entirely in raw signal space and does not produce discrete, reusable token representations, nor does it incorporate any latent-space self-supervised objective.

We argue that reconstruction in the raw signal space is a suboptimal pretraining objective for learning *semantic* EEG representations. First, raw EEG contains substantial low-level noise and physiological artifacts that do not carry task-relevant information; a model trained to reconstruct them may allocate significant capacity to irrelevant details. Second, patch-based masked reconstruction encourages local pattern matching rather than higher-level semantic structure. VAE-based approaches for EEG (Zhao et al., 2024; You et al., 2025) alleviate some of these issues by learning a compressed latent space, but they have not been scaled to large corpora or combined with structured self-supervised pretraining. In other domains—audio compression with EnCodec (Défossez et al., 2022) and image tokenization with RQ-VAE (Lee et al., 2022)—residual vector quantization has demonstrated that discrete, hierarchical codebooks provide compact yet expressive representations that benefit downstream modeling.

In this work, we introduce **EEG-RQVAE**, a foundation model for EEG based on a Residual Quantized Variational Autoencoder trained with a two-stage pretraining pipeline designed to learn increasingly semantic representations. Inspired by MedVAE (Varma et al., 2025), which showed that decoupling reconstruction quality from semantic structure through a dedicated second training stage substantially improves the informativeness of the latent space in medical imaging, we adopt a similar principle for EEG. In the first stage, the model is trained to reconstruct a clean, augmented view of the EEG from a corrupted input, leveraging denoising as an implicit regularizer while a patch discriminator enforces signal fidelity. In the second stage, we adapt the encoder using a Joint-Embedding Predictive Architecture (Assran et al., 2023) in latent space, predicting the representations of masked spatial and temporal regions from visible context—encouraging the model to capture the structured, predictive dependencies that underlie brain dynamics. Throughout both stages, we use bidirectional Mamba blocks as the backbone, enabling efficient modeling of long EEG sequences while respecting the temporal symmetry of brain signals.

Our main contributions are:

- **EEG-RQVAE**: a residual vector-quantized VAE for multi-channel EEG that produces discrete, hierarchical token sequences suitable for both generative and discriminative downstream use.
- **A two-stage pretraining pipeline**: combining denoising reconstruction (Stage 1) with latent JEPA-style adaptation (Stage 2) to progressively enforce semantic representation learning.
- **An analysis of SSM architectures for EEG**: we investigate the relevance of bidirectional Mamba blocks for spatio-temporal EEG modeling and compare with Transformer baselines.
- **A fair evaluation benchmark**: we evaluate learned representations with a standardized linear probing protocol across multiple BCI tasks, with a particular focus on imagined speech decoding, and compare against recent EEG foundation models under identical conditions.

2. Related Work

EEG foundation models. Building universal EEG representations has been approached through two main architectural families. Transformer-based models (Vaswani et al., 2017) have dominated early work: EEGPT (Wang et al., 2024) combines masked reconstruction with a spatio-temporal representation alignment objective, while CBraMod (Wang et al., 2025a) introduces a criss-cross attention mechanism to jointly model spatial and temporal dependencies across channels. REVE (Ouahidi et al., 2026) scales masked pretraining to 25,000 subjects to learn topology-agnostic representations, LUNA (Döner et al., 2025) proposes an efficient topology-agnostic masked pretraining approach, and CSBrain (Zhou et al., 2026) introduces a cross-scale spatiotemporal architecture to capture oscillations at multiple temporal resolutions. Despite strong performance under fine-tuning, Transformer-based models suffer from quadratic complexity in sequence length,

which limits their applicability to long-duration clinical recordings. State space models have emerged as a linear-time alternative: EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), and LuMamba (Broustail et al., 2026) all leverage Mamba-based backbones with masked reconstruction objectives. BAM-U-Net (Author, s) further integrates an adversarial patch discriminator into the denoising objective, yielding substantially stronger linear probing performance. However, all of these models share a common limitation: their pretraining objective operates in raw signal space, encouraging the model to faithfully reproduce waveform details that may be uninformative for downstream BCI tasks. Furthermore, none of them produce *discrete* token representations, which precludes their direct use in generative autoregressive pipelines.

Neural signal tokenization. Discrete representation learning has a long history in audio and image modeling. Residual Vector Quantization (RVQ) refines standard vector quantization by sequentially quantizing the residual error across multiple codebooks, achieving a favorable rate–distortion trade-off without increasing individual codebook size. EnCodec (Défossez et al., 2022) demonstrated that RVQ combined with adversarial training yields state-of-the-art audio compression, while RQ-VAE (Lee et al., 2022) showed that residual quantization of image features enables high-fidelity autoregressive image generation. In the EEG domain, LaBraM (Jiang et al., 2024) introduced a neural tokenizer based on vector-quantized spectral prediction, producing neural codes that serve as pretraining targets for a Transformer encoder. EEG-RQVAE extends this direction by combining RVQ with a two-stage pretraining pipeline applied directly to raw waveforms, without relying on pre-extracted spectral features.

Latent self-supervised learning and semantic representations. A growing body of work argues that self-supervised objectives formulated in latent space, rather than in signal space, yield more semantic representations. The Joint-Embedding Predictive Architecture (JEPA) (Assran et al., 2023) formalizes this intuition: instead of reconstructing masked patches from pixels, a predictor network learns to anticipate the latent representations of masked regions from visible context, preventing the model from expending capacity on low-level signal details. In the medical imaging domain, MedVAE (Varma et al., 2025) demonstrated that decoupling reconstruction quality from semantic structure through a dedicated second training stage—where the latent space is further refined using a vision-language model as a semantic anchor—significantly improves the utility of frozen representations for downstream classification. VAE-based approaches for EEG (Zhao et al., 2024; You et al., 2025) have confirmed that a compressed latent space captures clinically relevant features more compactly than raw signals, but these models have not been scaled to large multi-task corpora and their latent spaces are not shaped by any semantic self-supervised objective. EEG-RQVAE bridges these lines of work by (i) learning a discrete, hierarchically quantized latent space via RVQ, (ii) enforcing signal-level fidelity through adversarial denoising in Stage 1, and (iii) refining the encoder with a JEPA-style objective in Stage 2 that pushes representations toward semantic predictability without any reconstruction pressure.

3. Method

EEG-RQVAE is a foundation model for multi-channel EEG that learns discrete, hierarchically quantized representations through a two-stage pretraining pipeline. Figure 1 gives an overview of the full system. In Stage 1, an encoder–RVQ–decoder architecture is trained end-to-end to reconstruct a clean EEG window from a corrupted input, with an adversarial patch discriminator enforcing signal fidelity beyond what a pointwise loss can capture. In Stage 2, the decoder is frozen and the encoder is further adapted with a JEPA-style objective that operates entirely in discrete latent space, steering representations toward semantic predictability without any signal-level reconstruction pressure. Both stages share the same bidirectional Mamba-2 backbone. We first introduce the building blocks on which EEG-RQVAE relies, then describe the architecture and each training stage in detail.

3.1. Preliminaries

Selective state space models and bidirectional Mamba. State space models (SSMs) define a linear recurrence over a continuous input signal $u(t)$:

$$h'(t) = \mathbf{A} h(t) + \mathbf{B} u(t), \quad y(t) = \mathbf{C} h(t), \quad (1)$$

where $h(t) \in \mathbb{R}^N$ is the latent state and $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are learned parameters. After zero-order-hold discretization with step size Δ , the recurrence becomes:

$$h_t = \bar{\mathbf{A}} h_{t-1} + \bar{\mathbf{B}}_t x_t, \quad y_t = \mathbf{C}_t h_t, \quad (2)$$

with $\bar{\mathbf{A}} = e^{\Delta \mathbf{A}}$ and $\bar{\mathbf{B}}_t = \Delta \mathbf{B}_t$. Mamba (Gu & Dao, 2024) and its successor Mamba-2 (Dao & Gu, 2024) make $\mathbf{B}_t$, $\mathbf{C}_t$, and Δ input-dependent, yielding a *selective* SSM that can focus on or filter out individual time steps. Crucially, the

recurrence in Eq. (2) is computed with a hardware-efficient parallel scan, achieving *linear* complexity $\mathcal{O}(T)$ in the sequence length T , in contrast to the $\mathcal{O}(T^2)$ cost of self-attention.

Because EEG windows are processed offline rather than as causal streams, future context is as informative as past context. We therefore adopt a bidirectional variant (Author, s) that processes the sequence in both directions and fuses the two outputs by summation. Denoting by $\mathcal{M}(\cdot)$ a single pre-normalized residual Mamba-2 block,

$$\mathcal{M}(X) = \text{Mamba2}(\text{LN}(X)) + X, \quad (3)$$

the BiMamba-2 block with sum fusion reads:

$$\text{BiMamba}(X) = \mathcal{M}_{\text{fwd}}(X) + \text{rev}(\mathcal{M}_{\text{bwd}}(\text{rev}(X))), \quad (4)$$

where $\text{rev}(\cdot)$ reverses the sequence along the time axis. Each representation thus attends to both past and future temporal context while retaining linear-time inference.

Residual Vector Quantization. Vector Quantization (VQ) maps a continuous vector $z \in \mathbb{R}^d$ to its nearest entry in a finite codebook $\mathcal{E} = \{e_k\}_{k=1}^K \subset \mathbb{R}^d$:

$$q = \arg \min_k \|z - e_k\|_2. \quad (5)$$

Standard VQ is limited by the trade-off between codebook size K and representation precision. Residual Vector Quantization (RVQ) (Défossez et al., 2022; Lee et al., 2022) overcomes this by applying N quantization steps in cascade: the first codebook $\mathcal{E}^{(1)}$ quantizes the encoder output $z^{(0)} = z_e$, and each subsequent codebook $\mathcal{E}^{(n)}$ quantizes the residual left by the previous step:

$$q^{(n)} = \arg \min_k \|z^{(n-1)} - e_k^{(n)}\|_2, \quad z^{(n)} = z^{(n-1)} - e_{q^{(n)}}^{(n)}, \quad (6)$$

yielding the quantized representation $z_q = \sum_{n=1}^N e_{q^{(n)}}^{(n)}$. Gradients are propagated through the quantization bottleneck via the straight-through estimator (Défossez et al., 2022), and the encoder is regularized by a commitment loss:

$$\mathcal{L}_{\text{commit}} = \|z_e - \text{sg}[z_q]\|_2^2, \quad (7)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator. RVQ thus produces a sequence of N discrete indices per time step, forming a compact hierarchical token representation.

Joint-Embedding Predictive Architecture. JEPA (Assran et al., 2023) is a self-supervised learning framework that avoids signal-level reconstruction entirely. Given an input x , two views are created: a context x_c (visible patches) and a target x_t (masked patches). The context is encoded by an online encoder f_θ , and a lightweight predictor g_ϕ maps the context representation to a prediction of the target representation:

$$\hat{s}_t = g_\phi(f_\theta(x_c), m), \quad (8)$$

where m encodes the spatial position of the masked region. The target representation $s_t = f_\theta(x_t)$ is produced by an exponential moving average (EMA) copy of the encoder with parameters $\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta$, which provides a stable prediction target without requiring negative samples. The training objective minimizes the ℓ_2 distance between prediction and target in representation space:

$$\mathcal{L}_{\text{JEPA}} = \|\hat{s}_t - \text{sg}[s_t]\|_2^2. \quad (9)$$

Because the loss is computed entirely in latent space, the model is never penalized for ignoring high-frequency signal details irrelevant to the prediction task, which biases learning toward structured, semantic features.

3.2. EEG-RQVAE Architecture

Figure 1 gives an overview of EEG-RQVAE. A raw EEG window $x \in \mathbb{R}^{C \times T}$ ($C = 19$ channels, T time samples) is processed by four successive components: a channel adaptor, a spatio-temporal mixing block, a hierarchical convolutional encoder, and a residual vector quantizer. A symmetric decoder is used during Stage 1 only and is frozen for Stage 2.

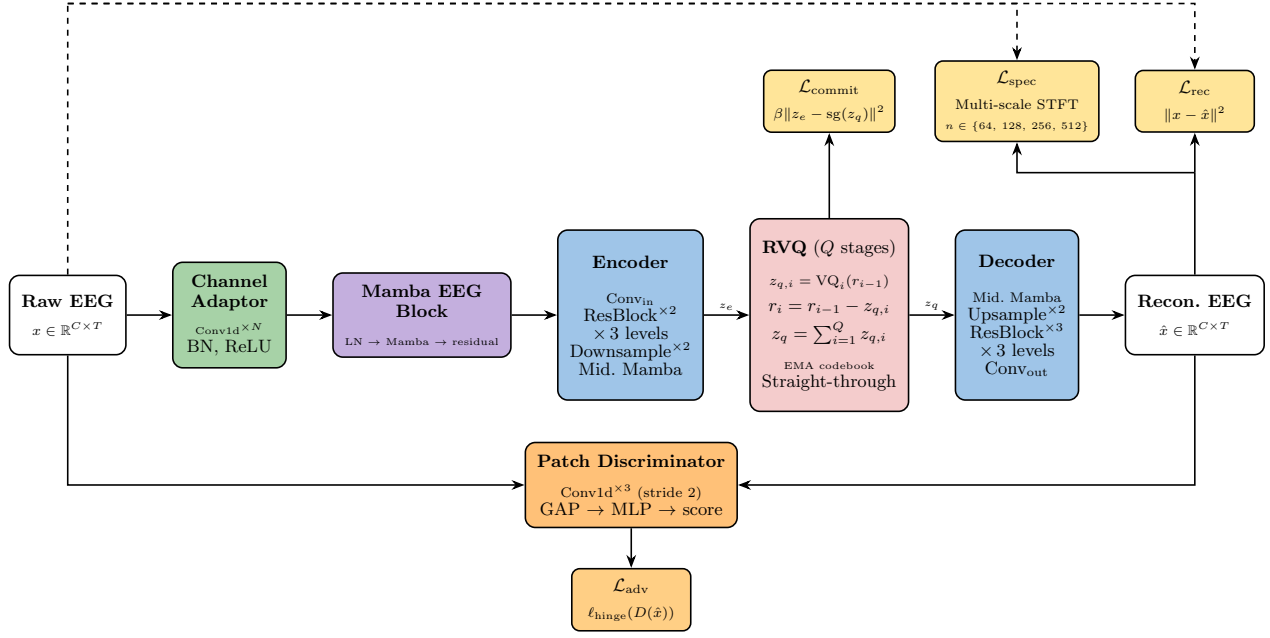


Figure 1. Overview of the EEG-RQVAE architecture. A raw EEG signal is first projected by a channel adaptor, then processed by a Mamba EEG Block that captures long-range temporal dependencies before downsampling. A hierarchical encoder maps the sequence to a compact latent code z_e , which is discretized by the residual vector quantizer (RVQ, Q stages) into z_q . A symmetric decoder reconstructs the signal from z_q (Stage 1 only). Losses are colour-coded: reconstruction $\mathcal{L}_{\text{rec}}$, multi-scale spectral $\mathcal{L}_{\text{spec}}$, commitment $\mathcal{L}_{\text{commit}}$, and adversarial $\mathcal{L}_{\text{adv}}$.

Channel adaptor. EEG recordings vary in electrode count and montage across devices and datasets. To decouple the model’s internal width from the raw channel count, we first pass the signal through a lightweight channel adaptor consisting of $L = 2$ point-wise convolutions (CONV1d, kernel size 1) interleaved with batch normalization and ReLU activations:

$$x' = \phi_{\text{adapt}}(x) \in \mathbb{R}^{C' \times T}, \quad C' = 64. \quad (10)$$

Because all convolutions have kernel size 1, the adaptor mixes channels but does not modify the temporal dimension.

Mamba EEG mixing block. Before entering the hierarchical encoder, x' passes through a *Mamba EEG Block* that captures global temporal dependencies across all time steps in a single selective-SSM pass. The block applies group normalization, a unidirectional Mamba scan, and a residual connection:

$$\text{MambaEEG}(x') = x' + \text{Mamba}(\text{GN}(x')), \quad (11)$$

where the Mamba scan operates along the time axis on the transposed sequence (B, T, C') and the output is transposed back to (B, C', T) . This lightweight pre-encoder stage mixes information across the full window length before the strided convolutions irreversibly reduce temporal resolution, enabling the encoder to leverage long-range context that would otherwise be truncated by early downsampling. An alternative attention-based variant (*CrossCross EEG Attention*) that factorizes spatial and temporal attention at the patch level with an additional FFT spectral branch is also supported but was not used in the experiments reported here.

Hierarchical encoder. The output of the mixing block is projected to $d_0 = 128$ feature maps by a CONV1d with kernel size 3, then processed by three resolution levels. At each level $\ell \in \{0, 1, 2\}$, two residual blocks expand the channel width:

$$d_\ell = 128 \times [1, 2, 4]_\ell = \{128, 256, 512\}, \quad (12)$$

and a strided convolution (stride 2) halves the temporal resolution at the first two levels. The signal is therefore downsampled by a factor of 4 along the time axis ($T \rightarrow T/4$). A bottleneck follows the last resolution level: two residual blocks

sandwich a Mamba EEG Block, providing global temporal context at the coarsest resolution. A final `Conv1d` projects to $z_e \in \mathbb{R}^{D \times (T/4)}$ with $D = 16$ via a `quant_conv` layer (kernel size 1).

Each residual block applies group normalization, a Swish activation, and two `Conv1d` layers (kernel size 3, padding 1) with a skip connection.

Residual vector quantizer. The encoder output z_e is passed to an RVQ with $N = 8$ stages, each stage using an EMA-updated codebook of $K = 512$ entries of dimension $D = 16$. The cascade produces the quantized representation $z_q = \sum_{n=1}^N e_{q(n)}^{(n)} \in \mathbb{R}^{D \times (T/4)}$ and a set of N discrete index sequences $\mathbf{q} \in \{1, \dots, K\}^{N \times (T/4)}$ that constitute the compact token representation of the EEG window. A `post_quant_conv` (kernel size 1) projects z_q before the decoder. The effective vocabulary over a single time step is $K^N = 512^8$, while each individual codebook lookup remains tractable. Codebook entries not used in a batch are replaced by a randomly sampled encoder output to prevent collapse (Laplace smoothing is applied to cluster sizes).

Decoder. The decoder mirrors the encoder: a bottleneck (residual blocks with sequence block) is followed by three resolution levels that progressively double the temporal resolution via nearest-neighbor upsampling and `Conv1d`, reaching the original length T . Each level uses `num_res_blocks` + 1 = 3 residual blocks. A final `Conv1d` maps to $\hat{x} \in \mathbb{R}^{19 \times T}$. The decoder is trained jointly with the encoder in Stage 1 and frozen thereafter; only the encoder (and its channel adaptor) is retained for downstream probing.

3.3. Stage 1 — Denoising Reconstruction

The objective of Stage 1 is to train the full encoder–RVQ–decoder pipeline to produce physiologically plausible reconstructions of clean EEG from corrupted inputs, while building a compact discrete latent space.

Data augmentation and corruption. Each training sample undergoes two independent stochastic transformations. First, a set of *augmentations* is applied to produce the *clean target* x_{tgt} : a random circular time shift (up to 10% of the window length), multiplicative amplitude scaling drawn from $\mathcal{U}(0.8, 1.2)$, dropout of up to 2 randomly selected channels (set to zero), additive Gaussian noise ($\sigma = 0.01$), and contiguous time masking of 5–10% of the sequence. Second, the augmented signal x_{tgt} is *corrupted* to produce the model input x_{in} : additional channel dropout and Gaussian noise are applied independently from the augmentation step, so that the model must recover meaningful structure rather than simply inverting a known transformation. This double-perturbation strategy encourages the encoder to learn representations that are invariant to the acquisition-level artefacts most commonly encountered in EEG, such as electrode impedance fluctuations and movement noise, while the decoder is trained to restore clean waveform morphology.

Reconstruction and spectral losses. The generator is trained to minimize a combination of a waveform loss and a multi-scale spectral loss between the reconstruction $\hat{x}$ and the *clean target* x_{tgt} (not the corrupted input x_{in}):

$$\mathcal{L}_{\text{rec}} = \text{SmoothL1}(\hat{x}, x_{\text{tgt}}), \quad (13)$$

$$\mathcal{L}_{\text{spec}} = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \mathcal{L}_{\text{L1}}(|\text{STFT}_n(\hat{x})|, |\text{STFT}_n(x_{\text{tgt}})|), \quad (14)$$

where $\mathcal{N} = \{64, 128, 256, 512\}$ are FFT window sizes (corresponding to 320 ms, 640 ms, 1.28 s, and 2.56 s at 200 Hz), covering all clinically relevant EEG frequency bands from delta to gamma. The multi-scale STFT loss penalizes spectral discrepancies at multiple temporal resolutions and is essential for preserving rhythmic EEG features that a pointwise loss tends to smooth out.

Adversarial loss. Starting from epoch 5, a patch-based discriminator D (NLayer-1D, $n_{\text{df}} = 64$, 3 strided convolutional layers) is introduced to enforce local temporal realism. The discriminator receives both the real signal x_{tgt} and the reconstruction $\hat{x}$ and is trained with the hinge loss:

$$\mathcal{L}_D = \frac{1}{2} (\mathbb{E}[\text{ReLU}(1 - D(x_{\text{tgt}}))] + \mathbb{E}[\text{ReLU}(1 + D(\hat{x}))]). \quad (15)$$

The generator is adversarially updated with:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}[D(\hat{x})]. \quad (16)$$

The adversarial objective forces the generator to produce high-frequency EEG patterns—such as sleep spindles or gamma-band oscillations—that are physiologically plausible but are typically smoothed out by the ℓ_1 or ℓ_2 reconstruction losses alone.

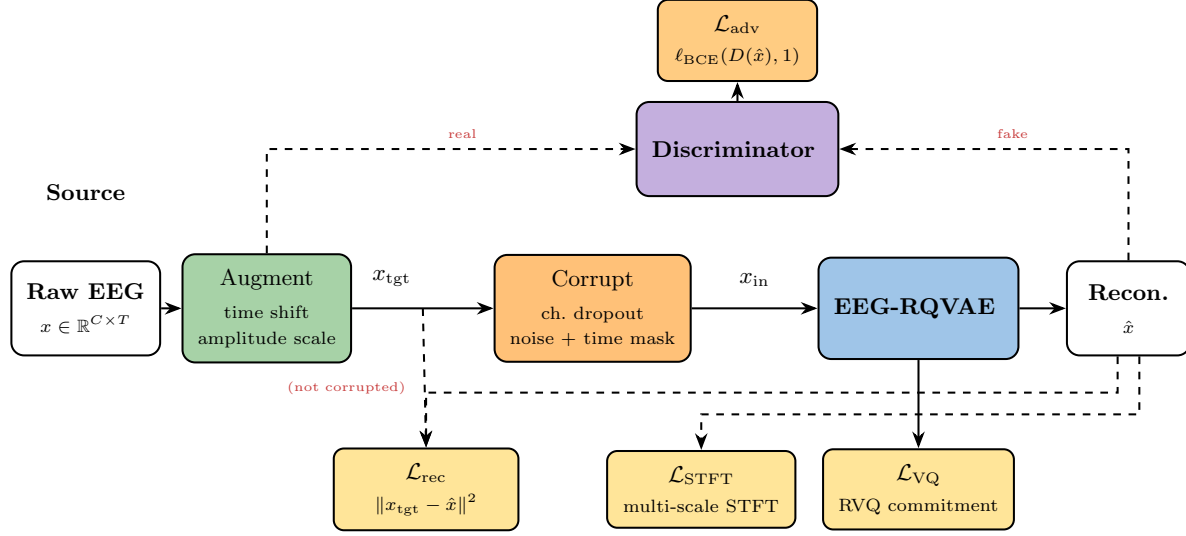
Total Stage 1 objective. The full generator loss combines the four terms:

$$\mathcal{L}_{\text{Stage 1}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} + \lambda_{\text{commit}} \mathcal{L}_{\text{commit}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}, \quad (17)$$

with $\lambda_{\text{spec}} = 0.1$, $\lambda_{\text{commit}} = 1.0$, and $\lambda_{\text{adv}} = 0.01$. The commitment loss $\mathcal{L}_{\text{commit}}$ is defined in Eq. (7) and regularizes the encoder output toward the RVQ codebook.

3.4. Stage 2 — Latent JEPA Adaptation

Stage 1: Reconstruction Pretraining



Stage 2: EEG-JEPA Adaptation

Encoder weights initialized from Stage 1

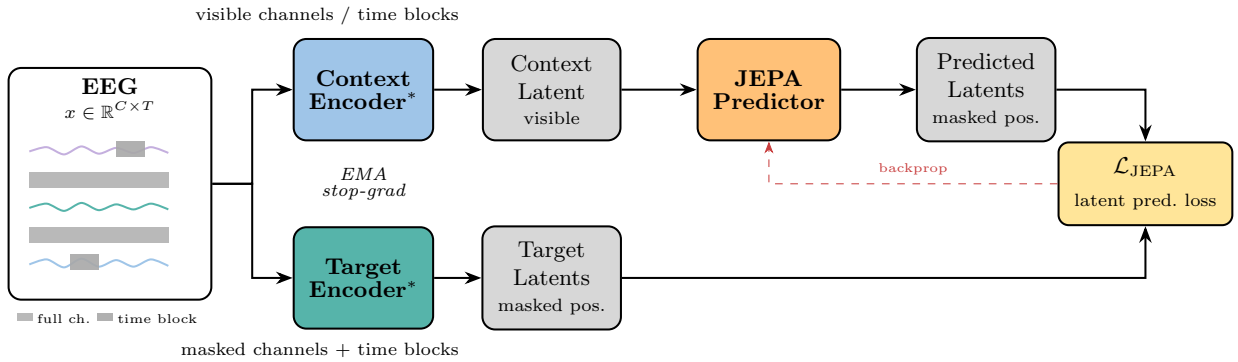


Figure 2. Two-stage pretraining pipeline. **Stage 1** (top): the raw EEG is first augmented to produce a clean target x_{tgt} , then independently corrupted to obtain the model input x_{in} . EEG-RQVAE reconstructs $\hat{x}$ from x_{in} and is trained to match x_{tgt} . **Stage 2** (bottom): the decoder is discarded; the encoder is adapted by a JEPA-style objective that predicts masked latent representations from visible context. * Encoder weights are initialized from Stage 1.

Stage 2 refines the encoder representations toward semantic predictability. The decoder and the RVQ codebook are frozen; only the encoder (channel adaptor, convolutional layers, and bottleneck) is updated. No reconstruction objective is used. This clean separation mirrors the strategy of MedVAE (Varma et al., 2025), which showed that decoupling reconstruction quality from semantic structure through a dedicated second stage substantially improves downstream discriminability.

Model. The Stage 2 model consists of three components: (i) a *context encoder* (initialized from Stage 1) that processes the observed EEG window; (ii) a *target encoder*, an exponential moving average (EMA) copy of the context encoder whose

parameters are never updated by gradient; (iii) a *narrow predictor* g_ϕ , a compact Transformer (4 blocks, $d_{\text{pred}} = 128$, 4 heads) that operates on the quantized latent sequence. The predictor first projects $z_q \in \mathbb{R}^{D \times T_z}$ from $D = 16$ to its internal dimension $d_{\text{pred}} = 128$, adds a learnable positional embedding, and applies 4 self-attention blocks before projecting back to $D = 16$. The “narrow predictor” design (Assran et al., 2023) is intentional: giving the predictor strictly less capacity than the encoder prevents it from trivially solving the prediction task without learning structure.

Latent masking. The temporal dimension of the quantized sequence $z_q \in \mathbb{R}^{D \times T_z}$ (where $T_z = T/4$) is divided into non-overlapping blocks of $B = 25$ tokens, corresponding to approximately 500 ms at 200 Hz with the encoder’s $4 \times$ temporal compression. At each training step, 25% of these blocks are randomly selected as the *target set* $\mathcal{T}$; the remaining blocks form the *context set* $\mathcal{C}$. Target positions in z_q are replaced by a shared learnable mask token $m \in \mathbb{R}^D$ (initialized from $\mathcal{N}(0, 0.02^2)$) before the context encoder, producing the masked sequence $\tilde{z}$. The target encoder always receives the complete, unmasked signal x .

JEPA objective. The predictor receives $\tilde{z}$ and the set of target positions $\mathcal{T}$ and produces predicted latent representations $\hat{s}_{\mathcal{T}} \in \mathbb{R}^{D \times |\mathcal{T}|}$. The target representations $s_{\mathcal{T}} = f_{\bar{\theta}}(x)_{\mathcal{T}}$ are extracted at the same positions by the EMA encoder. The Stage 2 loss is:

$$\mathcal{L}_{\text{Stage 2}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{s}_t - \text{sg}[s_t]\|_2^2, \quad (18)$$

where $\text{sg}[\cdot]$ denotes stop-gradient. After each gradient step, the target encoder is updated by EMA with a cosine-scheduled momentum that warms up from $\tau = 1.0$ to $\tau = 0.999$, following the approach of Assran et al. (2023):

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta. \quad (19)$$

The high final momentum ($\tau = 0.999$) provides a stable prediction target and avoids the representational collapse that can occur when context and target encoders converge too rapidly. Because the loss in Eq. (18) is computed entirely in the discrete latent space z_q , the model is never penalized for ignoring the low-level signal details already handled in Stage 1, biasing learning toward the structured temporal dependencies that underlie EEG cognitive dynamics.

4. Experiments

4.1. Datasets and Preprocessing

We pretrain on the Temple University Hospital EEG Corpus (TUEG) and evaluate the learned representations on twelve downstream BCI datasets spanning six task families: abnormal-EEG detection, sleep staging, emotion recognition, motor imagery, seizure detection, and imagined-speech decoding. Table 1 summarizes the corpora; full per-dataset details and label definitions are provided in Appendix C. All datasets are processed by a single, dataset-agnostic preprocessing pipeline so that pretraining and downstream signals share identical formatting and sampling rate.

Signal preprocessing. Each recording is loaded once with MNE, resampled to a common rate of 200 Hz, band-pass filtered to 0.5–75 Hz, and notch filtered at 60 Hz (all IIR filters). We then apply per-channel z-score normalization, computed over each window with a finite-value mask that neutralizes flat or non-finite channels. Class labels are read from each dataset’s native annotations and remapped to integer classes; segments without a valid label are discarded. Within every labeled segment we extract fixed-length windows whose duration is set per dataset to match the natural epoch of the task (the *Dur.* column of Table 1); when a segment does not divide evenly, the trailing window is anchored to the segment end to avoid zero-padding artifacts. For the unlabeled pretraining corpus, the entire recording is treated as a single segment and windowed with the same procedure. Each window is stored as an independent record in an LMDB store together with its subject identifier, sampling rate, channel names, and label, enabling scalable, subject-aware data loading.

Table 1. Datasets used for pretraining and downstream evaluation. Row 0 is used for pretraining only (no class labels). Rows 1–12 are evaluation datasets evaluated under linear probing.

BCI Task	Name	Ch.	Montage	Dur. (s)	Classes	Subjects	Samples
Pretraining (unlabeled)	TUH EEG Corpus	29	10-20 TCP (AR/LE)	30.0	—	1 385	303 686
Abnormal EEG detection	TUH Abnormal EEG	26	10-20 TCP (AR/LE)	30.0	2	1 164	68 486
Sleep staging	Sleep-EDF Expanded (50%)	2	Fpz-Cz / Pz-Oz	30.0	5	39	96 362
Emotion recognition	SEED-V	58	62ch 10-20	1.0	5	16	115 001
Motor imagery	PhysioNet EEG MMI	64	64ch 10-10	4.0	4	109	9 838
Emotion recognition	FACED	30	30ch 10-20	10.0	9	123	10 332
Motor imagery	BCI Competition IV-2a	22	22 EEG + 3 EOG	4.0	4	9	4 696
Seizure detection	Siena Scalp EEG	37	10-20	30.0	2	14	16 926
Seizure detection	CHB-MIT Scalp EEG	23	10-20 bipolar	10.0	1	24	353 872
Motor imagery	SHU-MI	32	32ch 10-20	4.0	2	25	11 988
Imagined speech (words)	BCI Competition 2020 T3	60	64ch 10-10	5.0	5	30	5 250
Imagined speech	KaraOne	58	64ch 10-20	5.0	11	14	1 913
Imagined speech	Chisco	62	122ch 10-10	4.0	39	5	32 403

4.2. Pretraining Setup

Pretraining corpus. Both stages are pretrained on the TUEG corpus, excluding the TUAB and TUAR subsets to prevent leakage into the downstream abnormal-EEG and artifact tasks. We sample 10% of the corpus files, yielding 1,385 subjects and 303,686 non-overlapping windows of 30 s at 200 Hz. The corpus is split at the subject level into 90% training and 5% validation (seed 42), so that no subject appears in both partitions.

Stage 1 — denoising reconstruction. The autoencoder is trained for 30 epochs with AdamW (learning rate 10^{-4} , weight decay 10^{-2} , $\beta = (0.5, 0.9)$), a cosine schedule, gradient clipping at 1.0, and mixed-precision training. The reconstruction objective is a smooth- ℓ_1 loss (weight 1.0) complemented by a multi-scale STFT spectral loss (weight 0.1; FFT window sizes {64, 128, 256, 512} samples, spanning 0.32–2.56 s at 200 Hz) and an adversarial term (weight 10^{-2}) provided by a patch discriminator. The discriminator (AdamW, learning rate 2×10^{-4}) is activated only after epoch 5 to let the autoencoder first stabilize, and is then updated jointly with the generator at every step. On the training split we apply stochastic EEG augmentations (each with probability 0.5): circular time shifts (up to 10% of the window), multiplicative amplitude scaling in $[0.8, 1.2]$, dropout of up to two channels, additive Gaussian noise ($\sigma = 0.01$), and contiguous time masking (5–10% of the window); the validation split is never augmented. The encoder maps the 19-channel input through a channel adaptor to 64 internal channels and a 3-level bidirectional Mamba backbone (base width 128, multipliers (1, 2, 4), two residual blocks per level) to a 16-dimensional latent. We train two variants of this stage that differ only in the latent bottleneck: a continuous **KL-Mamba** model regularized toward a unit Gaussian (KL weight 10^{-5}), and a residual vector-quantized **RVQ-Mamba** model with 8 quantizer stages, a codebook of 512 entries per stage (EMA decay 0.99), and commitment cost $\beta = 0.25$.

Stage 2 — latent JEPA adaptation. Starting from the frozen Stage 1 RVQ-Mamba checkpoint, the encoder is adapted with the latent JEPA objective for 30 epochs using AdamW (learning rate 2×10^{-5} , weight decay 5×10^{-2} , $\beta = (0.9, 0.95)$), a cosine schedule, gradient clipping at 1.0, and mixed precision. Targets are produced by an EMA copy of the encoder whose momentum follows a cosine warm-up from 1.0 to its final value 0.999. We predict 25% of the temporal blocks (block size 25 tokens, ≈ 500 ms given the $4\times$ temporal compression) from the visible context, using a lightweight Transformer predictor (embedding dimension 128, 4 heads, 4 layers) and a mean-squared-error loss in latent space; spatial channel masking is disabled in this configuration.

4.3. Pretraining Analysis

Before measuring downstream transfer, we analyze the pretrained models intrinsically, following the two-stage design: Stage 1 should faithfully reconstruct the EEG signal, while Stage 2 should add semantic structure to the latent space. All analyses below are computed on held-out PhysioNet MI windows that were never seen during pretraining.

Stage 1 reconstruction fidelity. Figure 3 overlays reconstructed and original waveforms for the three Stage 1 encoders across representative channels. All variants recover the dominant temporal morphology of the signal, including the oscillatory bursts and slow drifts, with the quantized models (RVQ-Mamba, RVQ-Transformer) tracking sharp transients slightly

more tightly than the continuous KL-Mamba. To verify that fidelity is not limited to the time domain, Figure 4 compares the average power spectral density (PSD) of the reconstructions against the original signal. The reconstructions match the original spectrum closely across the physiologically relevant δ - β bands (0.5–30 Hz); the RVQ models in particular preserve power up into the low- γ range, whereas the KL-Mamba model attenuates high-frequency content more aggressively, consistent with the smoothing induced by its Gaussian latent prior. The shared notch at 60 Hz reflects the line-noise filtering applied during preprocessing. Together these results confirm that Stage 1 yields high-fidelity reconstructions and that residual quantization does not degrade—and even slightly improves—spectral fidelity relative to the continuous bottleneck.

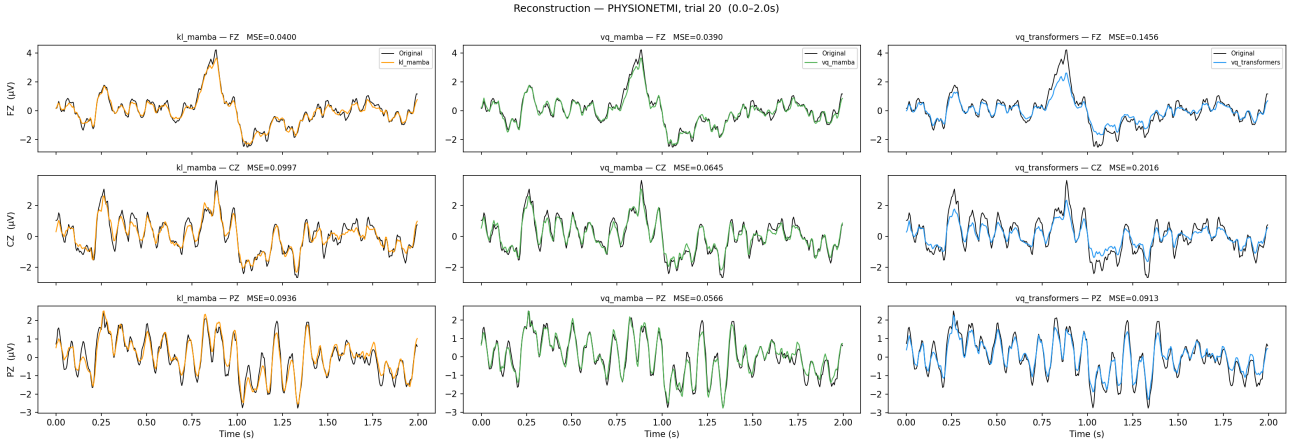


Figure 3. Stage 1 reconstruction in the time domain on held-out PhysioNet MI data. Original (solid) vs. reconstructed (dashed) waveforms for the KL-Mamba, RVQ-Mamba, and RVQ-Transformer encoders across representative channels.

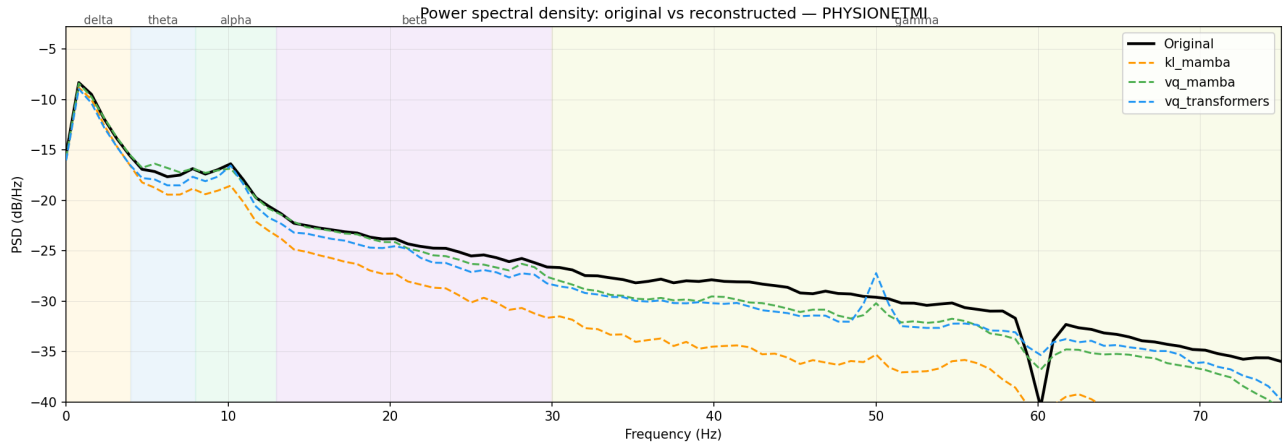


Figure 4. Stage 1 reconstruction in the frequency domain. Average power spectral density of the original signal and of each model’s reconstruction, with the canonical EEG bands ($\delta, \theta, \alpha, \beta, \gamma$) shaded. The RVQ variants preserve spectral power further into high frequencies than the continuous KL-Mamba model.

Stage 2 latent semantic structure. High reconstruction fidelity does not by itself imply that the latent space is organized along task-relevant axes. To probe this, we measure the k -nearest-neighbour label consistency in latent space ($k = 10$): the fraction of a window’s neighbours that share its class label, where a value of 0.25 corresponds to the random baseline for this four-class problem. Figure 5 reports this metric for the three Stage 1 encoders and for the Stage 2 JEPA model. The Stage 1 encoders—whether continuous or quantized—hover around the random baseline (0.243–0.251), indicating that reconstruction alone does not cluster the latent space by semantic class. The JEPA-adapted encoder is the only model to rise clearly above chance (0.266), showing that the latent prediction objective of Stage 2 instills semantic structure that pure signal reconstruction cannot. This corroborates our central hypothesis: predicting masked representations in latent space,

rather than reconstructing raw signal, is what drives the emergence of task-relevant structure, and it is consistent with the downstream gains reported in Section 4.4.

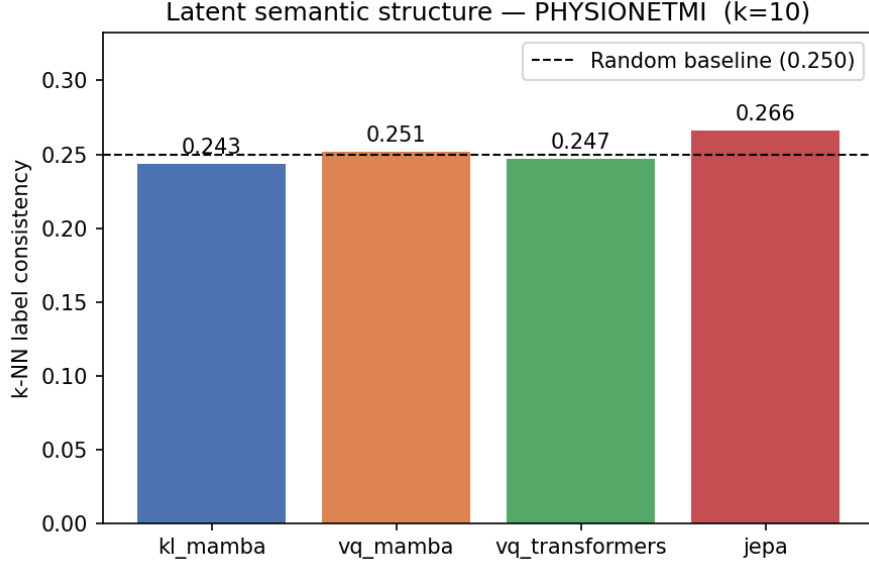


Figure 5. Latent semantic structure measured by k -NN label consistency ($k = 10$) on held-out PhysioNet MI data. Only the Stage 2 JEPA model rises clearly above the random baseline (dashed), whereas the Stage 1 encoders remain near chance.

4.4. Linear Probing Evaluation

Protocol. We evaluate the quality of the frozen representations with a standardized linear probing protocol: the pretrained backbone is kept frozen and only a lightweight linear head is trained on top of the pooled latent features for each downstream dataset. All datasets are processed by the unified pipeline of Section 4.1, and subjects are split at the subject level so the evaluation is fully subject-independent. We report balanced accuracy (BACC), Cohen’s κ , and macro-averaged F1 (MF1), each as mean $\pm$ std over multiple random seeds. We compare against recent EEG foundation models under identical conditions; baselines are evaluated with their publicly released checkpoints.

Main results. Among our four pretrained variants (compared in Table 5), the residual vector-quantized Mamba model (RVQ-Mamba) is the strongest overall and is therefore reported as our representative model in Tables 2 and 3. RVQ-Mamba is the best model on the motor-imagery dataset BCI2A across all three metrics, and it leads on the emotion (SEED-V), pathology (TUAB), and sleep-staging (Sleep-EDF) tasks, while remaining competitive on PhysioNet MI and FACED, where the attention-based EEGMamba and EEGPT baselines retain an edge. These results show that a frozen RVQ-Mamba encoder yields representations that transfer competitively across a broad range of BCI tasks.

Table 2. Linear probing results on the motor-imagery and emotion tasks (BCI2A, PhysioNetMI, FACED). Metrics reported: balanced accuracy (BACC), Cohen’s κ (KAPPA), and macro-averaged F1 (MF1). Baseline results are mean $\pm$ std over three seeds; EEG-VAE / RVQ-Mamba is averaged over six seeds of the band-pass-preprocessed runs. The per-architecture comparison of our four variants is reported in Table 5. **Bold:** best. Underline: second best.

Model	BCI2A			PhysioNetMI			FACED		
	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1
EEGPT (Wang et al., 2024)	0.347 $\pm$ 0.043	0.130 $\pm$ 0.058	0.341 $\pm$ 0.041	0.573 $\pm$ 0.023	0.430 $\pm$ 0.030	0.574 $\pm$ 0.024	0.413 $\pm$ 0.017	0.338 $\pm$ 0.021	0.412 $\pm$ 0.017
CBraMod (Wang et al., 2025a)	0.439 $\pm$ 0.084	0.252 $\pm$ 0.112	0.431 $\pm$ 0.079	0.472 $\pm$ 0.013	0.295 $\pm$ 0.017	0.470 $\pm$ 0.009	0.332 $\pm$ 0.009	0.247 $\pm$ 0.008	0.330 $\pm$ 0.007
EEGMamba (Wang et al., 2025b)	0.408 $\pm$ 0.083	0.210 $\pm$ 0.111	0.404 $\pm$ 0.083	0.578 $\pm$ 0.020	0.437 $\pm$ 0.026	0.576 $\pm$ 0.019	0.471 $\pm$ 0.007	0.402 $\pm$ 0.009	0.470 $\pm$ 0.008
REVE (Ouahidi et al., 2026)	—	—	—	—	—	—	—	—	—
CSBrain (Zhou et al., 2026)	—	—	—	—	—	—	—	—	—
LUNA (Döner et al., 2025)	—	—	—	—	—	—	—	—	—
LaBraM (Jiang et al., 2024)	—	—	—	—	—	—	—	—	—
BAM-U-Net (Author, s)	—	—	—	—	—	—	—	—	—
EEG-VAE / RVQ-Mamba (ours)	0.484 $\pm$ 0.128	0.313 $\pm$ 0.170	0.468 $\pm$ 0.124	0.519 $\pm$ 0.047	0.359 $\pm$ 0.062	0.518 $\pm$ 0.048	0.325 $\pm$ 0.017	0.241 $\pm$ 0.019	0.316 $\pm$ 0.018

Table 3. Linear probing results on the affective, pathology and sleep tasks (SEED-V, TUAB, Sleep-EDF). Metrics reported: balanced accuracy (BACC), Cohen’s κ (KAPPA), and macro-averaged F1 (MF1). Baselines are averaged over three seeds. For EEG-VAE/RVQ-Mamba, SEED-V is averaged over six seeds of the band-pass-preprocessed runs, whereas TUAB (three seeds) and Sleep-EDF (single seed) retain the earlier runs, which were not re-evaluated under the band-pass pipeline. **Bold**: best. Underline: second best.

Model	SEED-V			TUAB			Sleep-EDF		
	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1
EEGPT (Wang et al., 2024)	0.269 $\pm$ 0.058	0.083 $\pm$ 0.074	0.236 $\pm$ 0.087	<u>0.744</u> $\pm$ 0.014	<u>0.487</u> $\pm$ 0.034	<u>0.742</u> $\pm$ 0.016	<u>0.625</u> $\pm$ 0.020	<u>0.548</u> $\pm$ 0.028	0.586 $\pm$ 0.023
CBraMod (Wang et al., 2025a)	<u>0.278</u> $\pm$ 0.012	<u>0.099</u> $\pm$ 0.022	<u>0.250</u> $\pm$ 0.021	0.732 $\pm$ 0.011	0.465 $\pm$ 0.018	0.728 $\pm$ 0.013	0.457 $\pm$ 0.061	0.445 $\pm$ 0.047	0.453 $\pm$ 0.056
EEGMamba (Wang et al., 2025b)	0.257 $\pm$ 0.018	0.074 $\pm$ 0.020	0.221 $\pm$ 0.014	0.732 $\pm$ 0.008	0.463 $\pm$ 0.014	0.731 $\pm$ 0.007	0.584 $\pm$ 0.048	0.544 $\pm$ 0.034	<u>0.562</u> $\pm$ 0.041
REVE (Ouahidi et al., 2026)	—	—	—	—	—	—	—	—	—
CSBrain (Zhou et al., 2026)	—	—	—	—	—	—	—	—	—
LUNA (Döner et al., 2025)	—	—	—	—	—	—	—	—	—
LaBraM (Jiang et al., 2024)	—	—	—	—	—	—	—	—	—
BAM-U-Net (Author, s)	—	—	—	—	—	—	—	—	—
EEG-VAE/RVQ-Mamba (ours)	0.289 $\pm$ 0.036	0.110 $\pm$ 0.043	0.258 $\pm$ 0.045	0.767 $\pm$ 0.026	0.528 $\pm$ 0.057	0.762 $\pm$ 0.029	0.727 $\pm$ 0.000	0.592 $\pm$ 0.000	0.560 $\pm$ 0.000

Table 4. Frozen backbone size, latent representation shape, and number of trainable (linear probe) parameters per downstream dataset for each model variant.

Model	Frozen size (params)	Latent shape	Trainable parameters					
			BCI2A	PhysioNetMI	FACED	SEED-V	TUAB	Sleep-EDF
EEGPT (Wang et al., 2024)	25.3M	—	34.2K	35.0K	35.7K	35.2K	33.7K	34.1K
CBraMod (Wang et al., 2025a)	5.0M	—	17.7K	39.2K	92.8K	15.7K	42.9K	16.0K
EEGMamba (Wang et al., 2025b)	3.4M	—	17.7K	39.2K	92.8K	15.7K	42.9K	16.0K
REVE (Ouahidi et al., 2026)	—	—	—	—	—	—	—	—
CSBrain (Zhou et al., 2026)	—	—	—	—	—	—	—	—
LUNA (Döner et al., 2025)	—	—	—	—	—	—	—	—
LaBraM (Jiang et al., 2024)	—	—	—	—	—	—	—	—
BAM-U-Net (Author, s)	—	—	—	—	—	—	—	—
EEG-VAE/RVQ-Mamba (ours)	21.5M	—	186.6K	188.0K	187.5K	187.9K	186.3K	186.1K

4.5. Ablation Study

Encoder architecture and pretraining objective. We ablate the three design axes of EEG-RQVAE jointly by comparing the four pretrained variants under the same band-pass-preprocessed linear probing protocol and the same six seeds: the continuous **KL-Mamba** bottleneck (VAE without quantization), the residual vector-quantized **RVQ-Mamba** model, its Transformer-backbone counterpart **RVQ-Transformer** (which replaces the BiMamba sequence blocks by self-attention), and the **JEPA** model obtained after the Stage 2 latent adaptation of RVQ-Mamba. The comparison thus isolates (i) the effect of residual quantization (KL-Mamba vs. RVQ-Mamba), (ii) the effect of the SSM backbone (RVQ-Transformer vs. RVQ-Mamba), and (iii) the effect of the Stage 2 JEPA objective (RVQ-Mamba vs. JEPA). Table 5 reports the four datasets that were re-evaluated under the band-pass pipeline (BCI2A, PhysioNet MI, FACED, SEED-V).

Table 5. Architecture and pretraining-objective ablation under linear probing, on the four band-pass-preprocessed datasets. All variants share the frozen encoder size and probing protocol and are averaged over six seeds (mean $\pm$ std). **Bold**: best variant per row. Underline: second best.

Dataset	Metric	KL-Mamba	RVQ-Mamba	RVQ-Transformer	JEPA
BCI2A	BACC	0.456 $\pm$ 0.097	0.484 $\pm$ 0.128	<u>0.469</u> $\pm$ 0.099	0.443 $\pm$ 0.085
	κ	0.275 $\pm$ 0.130	0.313 $\pm$ 0.170	<u>0.292</u> $\pm$ 0.133	0.257 $\pm$ 0.114
	MF1	0.441 $\pm$ 0.099	0.468 $\pm$ 0.124	<u>0.451</u> $\pm$ 0.109	0.427 $\pm$ 0.090
PhysioNetMI	BACC	<u>0.536</u> $\pm$ 0.040	0.519 $\pm$ 0.047	0.528 $\pm$ 0.035	0.538 $\pm$ 0.026
	κ	<u>0.380</u> $\pm$ 0.053	0.359 $\pm$ 0.062	0.370 $\pm$ 0.046	0.384 $\pm$ 0.035
	MF1	<u>0.536</u> $\pm$ 0.040	0.518 $\pm$ 0.048	0.528 $\pm$ 0.034	0.544 $\pm$ 0.026
FACED	BACC	0.306 $\pm$ 0.016	0.325 $\pm$ 0.017	0.262 $\pm$ 0.008	<u>0.315</u> $\pm$ 0.010
	κ	0.220 $\pm$ 0.018	0.241 $\pm$ 0.019	0.172 $\pm$ 0.008	<u>0.230</u> $\pm$ 0.011
	MF1	0.298 $\pm$ 0.017	0.316 $\pm$ 0.018	0.250 $\pm$ 0.009	<u>0.307</u> $\pm$ 0.009
SEED-V	BACC	<u>0.287</u> $\pm$ 0.044	0.289 $\pm$ 0.036	0.280 $\pm$ 0.029	0.283 $\pm$ 0.027
	κ	<u>0.108</u> $\pm$ 0.057	0.110 $\pm$ 0.043	0.096 $\pm$ 0.033	0.104 $\pm$ 0.035
	MF1	0.258 $\pm$ 0.054	<u>0.258</u> $\pm$ 0.045	0.256 $\pm$ 0.035	0.255 $\pm$ 0.033

Three trends emerge. First, residual quantization helps on the dense emotion/motor-imagery datasets: RVQ-Mamba improves over the continuous KL-Mamba on BCI2A and FACED across all three metrics, confirming that the discrete codebook does not sacrifice—and in fact sharpens—the discriminability of the latent space. Second, the SSM backbone matters most on the high-channel emotion task: replacing BiMamba by self-attention (RVQ-Transformer) collapses FACED performance (MF1 0.250 vs. 0.316) while staying close elsewhere, indicating that the linear-time Mamba backbone captures spatio-temporal structure that the patch-attention variant misses on fine-grained affective classes. Third, the Stage 2 JEPA adaptation is task-dependent: it gives the best results on PhysioNet MI but slightly trails RVQ-Mamba on BCI2A, FACED and SEED-V, suggesting that latent prediction sharpens representations for some motor-imagery tasks without uniformly dominating. Overall RVQ-Mamba is the most consistent variant—best or second-best on every dataset—which motivates its use as our representative model in the main comparison of Section 4.4.

Channel mapping. Stage 1 pretraining is carried out on a fixed montage of 19 standard 10–20 electrodes. Downstream datasets, however, come with heterogeneous montages, so we compare two strategies for feeding their signals to the frozen encoder. In the *raw* regime we keep the native channels of the dataset and let a lightweight learned channel adapter project them onto the encoder input. In the *mapped* regime we instead reconstruct the pretraining layout: each of the 19 standard positions is filled with the spatially nearest available native channel, and any standard position without a corresponding electrode is set to zero. The mapped input thus matches exactly the channel layout the encoder saw during pretraining. Table 6 reports RVQ-Mamba under both regimes; TUAB is excluded as it is only available in the mapped form.

Table 6. Channel-mapping ablation for EEG-VAE/RVQ-Mamba: native channels with a learned adapter (*raw*) vs. projection onto the 19-channel pretraining montage with zero-filling (*mapped*). Mean $\pm$ std over three seeds; **bold** marks the better regime. [†]Sleep-EDF *raw* is a single seed (seed 8); SEED-V is omitted as no *raw* run is available.

Dataset	Raw (native channels)			Mapped (19-ch montage)		
	BACC	κ	MF1	BACC	κ	MF1
BCI2A	0.483 $\pm$ 0.140	0.309 $\pm$ 0.187	0.469 $\pm$ 0.151	0.495 $\pm$ 0.153	0.326 $\pm$ 0.204	0.475 $\pm$ 0.174
PhysioNetMI	0.565 $\pm$ 0.010	0.420 $\pm$ 0.014	0.564 $\pm$ 0.015	0.567 $\pm$ 0.003	0.422 $\pm$ 0.004	0.570 $\pm$ 0.003
FACED	0.346 $\pm$ 0.012	0.264 $\pm$ 0.012	0.342 $\pm$ 0.011	0.369 $\pm$ 0.014	0.289 $\pm$ 0.016	0.366 $\pm$ 0.016
Sleep-EDF [†]	0.727	0.592	0.560	0.200 $\pm$ 0.000	0.000 $\pm$ 0.000	0.093 $\pm$ 0.024

On the three dense motor/emotion datasets (BCI2A, PhysioNetMI, FACHED), mapping yields a small but consistent gain across all three metrics. When the native montage is rich enough to cover most of the standard 19 positions, reconstructing the pretraining layout lets the frozen encoder apply its learned spatial filters to electrodes at their expected locations, whereas the raw regime forces the adapter to re-learn this correspondence from few labels. The trend reverses sharply on Sleep-EDF: with only two EEG derivations, the mapping zero-fills 17 of the 19 inputs, the encoder receives an almost empty montage, and the probe degenerates to the majority class (BACC 0.20, $\kappa = 0$). Keeping the two native channels with a learned adapter is then essential and recovers strong performance (BACC 0.73). Channel mapping is therefore beneficial when the downstream montage is dense and close to the standard 10–20 system, but detrimental for sparse or atypical montages, where a learned adapter on the native channels should be preferred.

5. Limitations

Our study has several limitations that we make explicit.

Codebook stability. The residual quantizer relies on $N = 8$ EMA-updated codebooks. Although dead-code revival and Laplace smoothing mitigate collapse, the stability of the discrete codes remains sensitive to the size of the pretraining corpus and to the calibration of the latent dimension D . With a small or weakly diverse corpus, the higher residual stages risk under-utilization, which can cap the effective vocabulary well below its nominal K^N capacity.

Scale and diversity of pretraining. We pretrain on a single corpus (TUEG) and on only 10% of its files (1,385 subjects). Both the limited volume and the reliance on a single acquisition source likely under-estimate the potential of the learned representations: prior foundation-model work shows that corpus scale and montage/site diversity are primary drivers of transfer, so a larger, multi-corpus pretraining set could substantially improve generalization.

Comparability of baselines. Baselines are evaluated from their publicly released checkpoints rather than re-pretrained under our protocol. While the downstream linear-probing pipeline is identical for all models, the underlying pretraining data, corpus size, and optimization differ across methods; the comparison therefore reflects released models as-is rather than a strictly controlled, matched-pretraining study.

Unexplored loss interactions. The Stage 1 objective combines reconstruction, multi-scale spectral, commitment, and adversarial terms with fixed weights, and Stage 2 adds a latent prediction loss. We did not ablate the interaction between these terms, so the chosen weighting may be sub-optimal and the relative contribution of each loss to downstream quality remains uncharacterized.

Cost and uneven benefit of Stage 2. The latent JEPA stage roughly doubles the pretraining budget yet does not improve every task uniformly (cf. Table 5): it helps PhysioNet MI but slightly trails the Stage 1 RVQ-Mamba on BCI2A, FACHED and SEED-V. The semantic gain is therefore task-dependent and is not guaranteed for tasks with weaker semantic content.

Sequence directionality. While our framework is formulated with bidirectional Mamba blocks (Equation (4)), the checkpoints evaluated here use the unidirectional (causal) selective scan. Because EEG windows are processed offline, a strictly causal backbone discards future context that is as informative as past context; we therefore expect a fully bidirectional backbone to be better suited to EEG, and confirming this gain is left to future validation.

6. Future Work

Imagined-speech decoding. A central motivation for compact, semantic EEG tokens is communication BCIs. We plan a focused study of imagined-speech decoding on the BCI 2020 T3, KaraOne and Chisco benchmarks, including a systematic phoneme- versus word-level comparison, to assess the clinical relevance of the learned representations for speech-restoration interfaces.

Discrete tokens for generative modeling. EEG-RQVAE produces hierarchical discrete codes that are, by construction, suitable inputs for autoregressive sequence models. Coupling the frozen RVQ tokenizer with a decoder-only Transformer over the code sequence—analogueous to RQ-Transformer for images (Lee et al., 2022) and to neural audio language models—would enable conditional EEG generation, imputation, and forecasting.

Scaling. Given the limited corpus used here, characterizing how representation quality scales with the volume and diversity of pretraining data is a natural next step, and would clarify how far the present results are from the model’s ceiling.

Loss design. A systematic exploration of the Stage 1 loss weighting—the reconstruction/spectral/adversarial trade-off—together with the balance between Stage 1 and Stage 2, could yield better-calibrated representations than the fixed weights used here.

Semantically grounded and bidirectional adaptation. Two extensions to Stage 2 are promising: (i) driving the JEPA adaptation with task-relevant datasets to inject stronger semantic structure into the latent space, and (ii) replacing the causal backbone with the fully bidirectional Mamba blocks of Equation (4), which better match the offline, non-causal nature of EEG analysis.

Subject-specific modeling for imagined speech. A fundamental obstacle to EEG foundation models is the extreme inter-subject variability of brain signals: the spatial distribution of task-relevant activity, the dominant frequency bands, and the morphology of evoked responses all vary substantially across individuals, in ways that a single population-level model may fail to capture. This variability is particularly pronounced in imagined speech, where the neural correlates of phonological and lexical content are highly idiosyncratic. We argue that, rather than forcing a universal representation onto this diversity, a promising direction is to develop *subject-specific* models that explicitly exploit the structure of EEG for a given individual—leveraging the temporal regularities, spatial topography, and spectral fingerprints that are stable within a subject across sessions. Such models, fine-tuned or distilled from a foundation encoder, could yield representations that are far better calibrated to the signal geometry of a single brain, and would be better suited to the clinical reality of imagined-speech BCIs, where deployment is inherently single-user.

7. Conclusion

We presented EEG-RQVAE, a foundation model that recasts EEG pretraining as a two-stage process: adversarial denoising builds signal fidelity and a compact discrete codebook, while a subsequent latent JEPA stage steers the encoder toward semantic predictability without any reconstruction pressure. Our intrinsic analysis shows that this staged decoupling is necessary — reconstruction fidelity and latent semantic structure are orthogonal properties that a single-objective model cannot simultaneously optimise. Discrete residual quantization further provides a natural bridge between the encoder and the broader ecosystem of autoregressive sequence modeling, opening avenues for conditional EEG generation and imputation that continuous representations cannot easily support. We hope these findings encourage the community to move beyond pixel-space reconstruction as the default self-supervised objective for brain signals.

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A. Codebook Analysis

B. Foundation Models Used for Comparison

C. Dataset Details and Preprocessing

This appendix details the unified preprocessing pipeline and the dataset-specific label definitions summarized in Section 4.1 and Table 1.

C.1. Preprocessing Configuration

All datasets are processed with the single configuration reported in Table 7. The pipeline performs, in order: resampling to 200 Hz, IIR band-pass filtering (0.5–75 Hz), IIR notch filtering (60 Hz), per-channel z-score normalization, and fixed-length windowing. The window length is the only parameter varied across datasets and is set to the natural epoch of each task (Table 1); when a labeled segment is not an integer multiple of the window length, the last window is anchored to the end of the segment rather than zero-padded. Each resulting window is written as an independent LMDB record holding the signal tensor, sampling rate, channel names, subject identifier, and integer label.

Table 7. Shared preprocessing parameters.

Parameter	Value
Resampling rate	200 Hz
Band-pass filter	0.5–75 Hz (IIR)
Notch filter	60 Hz (IIR)
Normalization	per-channel z-score
Window length	per-dataset (see Table 1)
Window overlap	0 s
Trailing-window mode	anchored to segment end
Storage format	LMDB (one window per record)

D. Downstream Datasets Details

E. Ablation: Channel Mapping and Spatial Information

EEG datasets differ widely in montage, channel count, and naming conventions, so a foundation model trained on a fixed input layout requires a strategy to reconcile heterogeneous montages. We adopt a channel-mapping scheme that projects every dataset onto a common 19-channel subset of the international 10–20 system {FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Cz, Pz}, so that pretraining and all downstream tasks share an identical $19 \times T$ input.

Mapping procedure. Channel names are first normalized (uppercasing, stripping of EEG/REF/LE prefixes and suffixes, removal of separators) and matched against an explicit, hand-verified mapping table per dataset. Older nomenclature is remapped to its modern equivalent (e.g. T3→T7, T4→T8, T5→P7, T6→P8); high-density montages collapse neighbouring electrodes onto the nearest target site, and bipolar montages (e.g. CHB-MIT) are mapped from their derivation pairs. When several source channels map to the same target, midline and central electrodes are given priority. Non-EEG modalities (EOG, EMG, ECG, respiration, markers, mastoids, etc.) are removed. Target channels absent from a montage are zero-filled and flagged by a boolean presence mask, so the encoder always receives a $19 \times T$ tensor regardless of the source montage.

F. Computational Analysis

G. Implementation Details

H. Use of LLMs

Large language models (LLMs) were used to assist with writing and editing, including reformulating sentences and paragraphs for clarity. All model architectures, training pipelines, and evaluation code were written by the authors; LLM assistance was limited to time-consuming boilerplate components such as the data-loading parser.